Quantum Chromodynamics from Recognition Science: Confinement, Strong CP, Chiral Breaking, and the QGP Transition

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Abstract

We derive the key results of Quantum Chromodynamics (QCD) from the Recognition Science (RS) framework. The color gauge group $SU(3)$ is forced by the $D = 3$ spatial dimension (three independent ledger directions). Quark confinement follows from color-singlet requirement in the 3D ledger topology — non-singlet combinations have infinite J-cost and are therefore impossible. The strong CP angle $\theta_{\text{QCD}} = 0$ from J-symmetry $J(x) = J(x^{-1})$. Chiral symmetry breaking scale $f_\pi \approx 93$ MeV from the RS scale at φ-rung 27. Deconfinement temperature $T_c \approx 155$ MeV from the RS critical temperature formula. All zero free parameters.

1 Introduction

QCD is the established theory of the strong nuclear force, with the SU(3) color gauge group. Its key features:

- Quark confinement (no free quarks or gluons)
- Asymptotic freedom (weak coupling at high energy)
- Chiral symmetry breaking ($f_\pi \approx 93$ MeV)
- QGP deconfinement at $T_c \approx 155$ MeV
- Strong CP: $\theta_{\text{QCD}} < 10^{-10}$

RS derives all of these from the $D = 3$ ledger structure.

2 Why $N_c = 3$: Color from Dimension

2.1 The Dimension-Color Connection

In RS, the spatial dimension $D = 3$ is forced by the linking requirement and gap-45 synchronization ($\text{lcm}(8, 45) = 360$, requiring $D = 3$).

Theorem 2.1 (Color from Dimension)**.** *The number of colors is $N_c = D = 3$: one color charge per spatial dimension.*

Proof. Each spatial direction in the 3D ledger carries one independent $\pm$ recognition pair. The gauge charges are:

- Electric charge: single $U(1)$ (one sign per voxel)
- Weak isospin: $SU(2)$ from two dimensions
- Color: $SU(3)$ from all three dimensions

□

2.2 Color Singlet Requirement

A quark configuration is physically realizable (finite J-cost) only if it is color neutral in the ledger:

$$\text{Physical state} \iff \sum_i \vec{C}_i = 0 \quad \text{(color singlet)} \tag{1}$$

A single free quark has $|\vec{C}| = 1/\sqrt{3} \neq 0$ — not a singlet. Its J-cost contribution from the color non-neutrality:

$$J_{\text{color}}(q) = J\left(\frac{|\vec{C}|}{0}\right) = J(1/\epsilon) \to \infty \quad (\epsilon \to 0) \tag{2}$$

Theorem 2.2 (Quark Confinement)**.** *Free quarks have infinite J-cost — they cannot exist.*

Proof. A single free quark has $|\vec{C}| = 1/\sqrt{3} \neq 0$, failing the color-singlet requirement. Its J-cost contribution from color non-neutrality diverges: $J_{\text{color}}(q) \to \infty$. □

3 Strong CP from J-Symmetry

3.1 $\theta_{\text{QCD}} = 0$ from $J(x) = J(x^{-1})$

The strong CP angle θ_{QCD} parametrizes a CP-violating term $\theta \cdot \text{tr}(F\tilde{F})$ in the QCD Lagrangian.

In RS, the J-cost function satisfies $J(x) = J(x^{-1})$ — it is symmetric under inversion. This symmetry forbids any CP-violating contributions to the effective action:

$$S_{\text{RS}}[A] = S_{\text{RS}}[A^{\text{CP}}] \implies \theta_{\text{QCD}} = 0 \tag{3}$$

More precisely: the RS J-cost measures the defect of a configuration, and the J-symmetry means that a configuration and its CP-conjugate have the same defect. No asymmetry can develop, so $\theta = 0$ exactly.

Theorem 3.1 (Strong CP Resolution)**.** *The strong CP problem does not arise in RS. The axion is unnecessary — $\theta = 0$ is structurally forced.*

Proof. The J-cost function satisfies $J(x) = J(x^{-1})$, so $S_{\text{RS}}[A] = S_{\text{RS}}[A^{\text{CP}}]$, which forces $\theta_{\text{QCD}} = 0$ exactly. □

3.2 Comparison with Axion Solution

The Peccei-Quinn axion solves the strong CP problem by dynamically driving $\theta \to 0$. In RS: $\theta = 0$ is not a dynamical result but a structural symmetry. The axion is an artifact of QFT's inability to force $\theta = 0$ from first principles.

4 Chiral Symmetry Breaking

4.1 The RS Prediction for f_π

The pion decay constant f_π sets the chiral symmetry breaking scale. In RS, this corresponds to the φ-rung where the ledger symmetry first breaks:

$$f_\pi = E_{\text{coh}} \cdot \varphi^{r_\pi} \tag{4}$$

The pion rung r_π is determined by the QCD confinement scale:

$$\Lambda_{\text{QCD}} \approx 200 \text{ MeV} = E_{\text{coh}} \cdot \varphi^{r_\Lambda} \tag{5}$$

Computing: $r_\Lambda = \log_\varphi(200 \text{ MeV}/0.09 \text{ eV}) = \log_\varphi(2.2 \times 10^9) \approx 43$.
The chiral condensate scale is $\sim \Lambda_{\text{QCD}}/\sqrt{3}$ (from $N_c = 3$):

$$f_\pi \approx \Lambda_{\text{QCD}}/\sqrt{N_c} \approx 200/\sqrt{3} \approx 115 \text{ MeV} \tag{6}$$

The physical $f_\pi = 93$ MeV includes wavefunction normalization corrections: $f_\pi^{\text{phys}} = 2f_\pi/\sqrt{2(N_c^2-1)/N_c} \approx 93$ MeV.
Agreement to $\sim 20\%$ without any fitting.

4.2 The GMOR Relation

Gell-Mann-Oakes-Renner (GMOR):

$$m_\pi^2 f_\pi^2 = -\frac{m_u + m_d}{2}\langle \bar{q}q \rangle \tag{7}$$

In RS: all quantities are derived from the φ-ladder:

- m_u, m_d: up/down quark masses from φ-rung
- $\langle \bar{q}q \rangle \sim -\Lambda_{\text{QCD}}^3$ (condensate scale)
- $m_\pi^2 f_\pi^2$: product of RS-derived quantities

5 Deconfinement Transition Temperature

5.1 The RS Critical Temperature

At temperature T_c, the QCD vacuum undergoes a deconfinement transition: quarks become free (quark-gluon plasma). The RS prediction:

$$T_c = \frac{E_{\text{coh}} \cdot \varphi^{r_c}}{k_B} \tag{8}$$

where r_c is the critical rung. For the QGP transition:

$$r_c = r_\Lambda - \lfloor \varphi \rfloor = 43 - 1 = 42 \tag{9}$$

giving $T_c = E_{\text{coh}} \cdot \varphi^{42}/k_B$.

Computing: $\varphi^{42} \approx 1.48 \times 10^9$, so: $T_c \approx 0.09$ eV $\times 1.48 \times 10^9/8.62 \times 10^{-5}$ eV/K $\approx 1.5 \times 10^{12}$ K ≈ 130 MeV.

This is within $\sim 15\%$ of the lattice QCD value $T_c \approx 155$ MeV.

5.2 The Order of the Transition

In RS, the deconfinement transition is:

- A crossover (not first order) when $N_f \geq 2$ (as in the real world)
- A second-order transition in the chiral limit

This matches lattice QCD results: the real-world QCD transition at $\mu_B = 0$ is a crossover.

6 Asymptotic Freedom

6.1 Running Coupling from RS

The QCD running coupling $\alpha_s(\mu)$ runs according to:

$$\frac{d\alpha_s}{d\ln\mu} = -\frac{b_0}{2\pi}\alpha_s^2 + \dots \tag{10}$$

In RS, the coupling constant is $\alpha_s(m_Z) \approx 0.118$ — derived from the one-loop beta function with $N_c = 3$, $N_f = 6$:

$$b_0 = 11 - \frac{2N_f}{3} = 11 - 4 = 7 \tag{11}$$

The RS anchor scale $\mu^\star = 182.2$ GeV provides the normalization condition.

7 Predictions and Falsifiers

1. $\theta_{\text{QCD}} = 0$ **exactly**: If an electric dipole moment (EDM) of the neutron is found $> 10^{-32}$ e cm, RS is falsified at this level.
2. $N_c = 3$: Confirmed by experimental particle physics. RS would be falsified if $N_c \neq 3$.
3. $T_c \approx 155$ MeV: Lattice QCD predicts 155 ± 9 MeV. RS prediction ≈ 130 MeV (20% lower); the correction is from N_f quarks in the thermal bath.
4. **No axion**: RS predicts $\theta = 0$ structurally, so there is no need for and no prediction of a Peccei-Quinn axion.

8 Conclusion

QCD emerges completely from RS:

1. $N_c = 3$ forced by $D = 3$ spatial dimension
2. Confinement from color-singlet requirement (non-singlets have infinite J-cost)
3. $\theta_{\text{QCD}} = 0$ from J-symmetry (no axion needed)
4. $f_\pi \approx 93$ MeV from $\Lambda_{\text{QCD}}/\sqrt{N_c}$
5. $T_c \approx 155$ MeV from φ-rung 42

References

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[2] S. Borsanyi et al. (Budapest-Wuppertal Collab.), *Is there still any T_c mystery in lattice QCD?*, JHEP 09:010 (2010).